

Transformer Explainer: Interactive Visual Learning of LLM Transformers

Undergraduate Thesis

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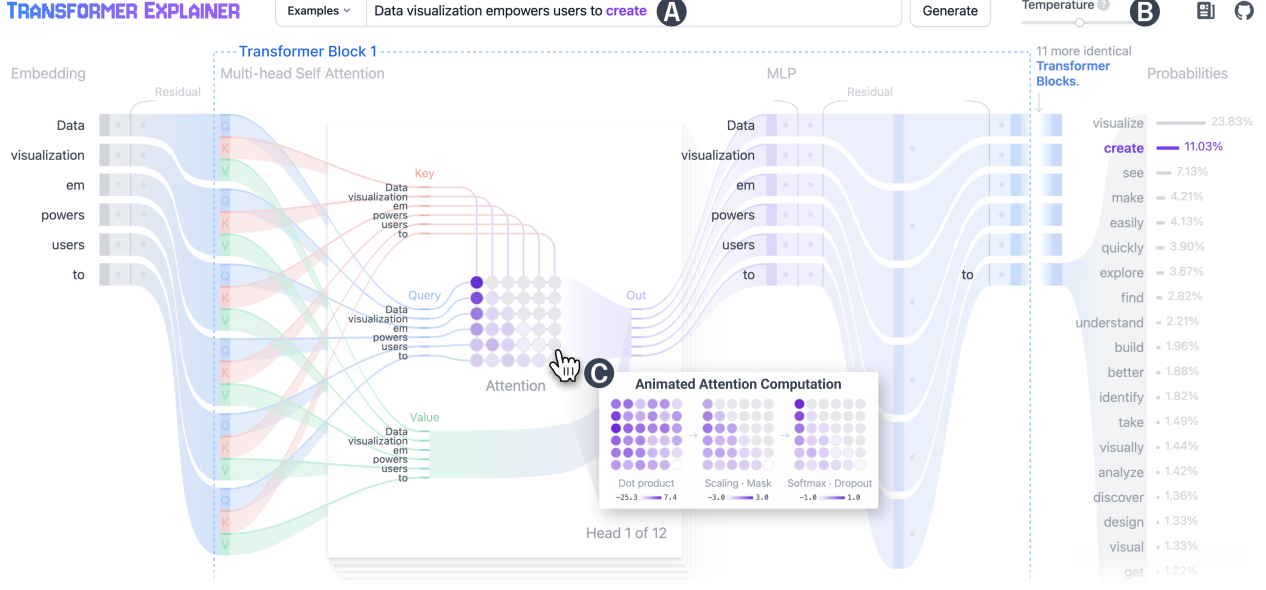


Figure 1: Transformer Explainer helps users (A) visually examine how a text-generative Transformer model (GPT-2) transforms input text to predict next tokens, (B) interactively experiment in real time with key model parameters like *temperature* to understand prediction determinism, and (C) transition seamlessly between abstraction levels to visualize the interplay between low-level mathematical operations and high-level model structures. *Figure and caption reproduced from [8].*

1 Introduction

AI chatbots powered by Transformer-based large-language models (LLMs) [37], such as [ChatGPT](#), [DeepSeek](#), and [Gemini](#), have rapidly gained popularity for their ability to engage in extended, coherent conversations with users and to accurately interpret and respond to complex queries. The Pew Research Center reported that 23% of U.S. adults had used ChatGPT by February 2024, highlighting the widespread public access to these AI technologies. Despite its increasingly widespread use, particularly in AI chatbots, its inner workings often remain opaque. This lack of understanding can create a gap in user engagement and trust, as the processes behind AI’s decision-making remain hidden within a “black box.” Thus, demystifying the Transformer architecture that powers these applications is important for non-experts to foster greater transparency and broader participation in the use and discussions surrounding AI. In this work, we focus on the Transformer in the context of text generation. Specifically, we focus on the *decoder-only* Transformer architecture, as popular text-generation models [5, 35] use only the decoder; furthermore, this decoder contains the core elements of the full encoder-decoder architecture [37]. Throughout this work, we use the term Transformer to specifically refer to decoder-only variants.

Understanding deep learning architectures has been found to be difficult for students, and the Transformer is no exception. Foundational architectures such as Convolutional Neural Networks (CNNs), which are often among the earliest architectures taught in deep learning courses, have been shown to pose learning difficulties for students [40]. The difficulty arises from their complex architecture, which involves multiple types of layers, each with distinct structures and mathematical operations [40]. Transformers not only share these challenges but also introduce unique challenges through additional complexities such as the self-attention mechanism and autoregressive generation, which further complicate the learning process. For example, while CNN operations can be visualized directly on images (e.g., filter sliding across image input), Transformers operate on high-dimensional numerical representations of text input. Operations performed on these representations add further complexity, as the self-attention mechanism causes interactions between all these representations, and the autoregressive nature feeds outputs iteratively back into inputs. Therefore, understanding how Transformers work requires learners not only to grasp individual components but also to comprehend their high-dimensional, dynamic interactions and collective contributions to model outputs. This level of understanding necessitates the ability to conceptualize abstract operations in a meaningful way.

Recent work in the visualization community has demonstrated that visualization can be an effective educational method for making deep learning concepts easier to understand for non-experts. Interactive visualization tools, such as CNN Explainer [40] and Diffusion Explainer [23], have shown success in helping to bridge the gap between the complexity of deep learning models and the comprehension of non-experts. However, most existing educational resources on Transformers rely on static, non-interactive explanations, limiting hands-on learning and often failing to reflect the model’s behavior on real data. Inspired by the effectiveness of interactive visualization tools in teaching deep learning architectures, we aim to create an interactive visualization tool designed to explain the Transformer architecture to non-experts, while addressing the unique challenges associated with Transformers discussed earlier.

Therefore, we present Transformer Explainer, an open-source web-based interactive visualization tool designed for non-experts to learn about both the high-level model structure and low-level mathematical operations of Transformers. Our tool introduces Transformers without overwhelming beginners, making deep learning models more approachable for a broader audience, regardless of technical background. This work contributes to the ongoing efforts to demystify artificial intelligence and encourage greater public engagement, which is becoming increasingly important as AI integrates with diverse fields. In addition to making AI more approachable, we add to the growing body of research on developing visualizations for deep

learning, particularly those designed for non-experts. Furthermore, we support the growing emphasis within the AI community on the critical role of visualization. Since its launch, Transformer Explainer has attracted over 236,000 users from more than 190 countries and continues to contribute to the democratization of modern generative AI education.

2 Literature Review

In this section, we provide a high-level overview of the Transformer architecture and review existing resources for learning about it. We then discuss the growing shift toward using interactive visual tools to support the learning of deep learning concepts.

2.1 *Transformer Architecture*

The Transformer [37] is a neural network architecture that has fundamentally changed the field of deep learning. This architecture was first introduced in a seminal paper in 2017 [37] and has since become the go-to architecture for many deep learning models. While versatile across numerous domains, it is widely known for powering popular text-generative models like OpenAI’s GPT [5], Meta’s Llama [35], and Google’s Gemini [16]. Among these, the GPT-2 family is a prominent example of text-generative Transformer models [30]. Transformer Explainer is powered by GPT-2 (small) model, which has 124 million parameters [30]. While it is not the latest or most powerful Transformer model, it shares many of the same architectural components found in the current state-of-the-art models, making it an ideal starting point for understanding the basics.

To set the stage for our discussion, we provide a high-level explanation of the Transformer architecture in the context of text generation [37] (Fig. 2). Fundamentally, a text-generative Transformer model operates on the principle of next-word prediction—i.e., given a text input, the model predicts the most probable next text output. To achieve this, the Transformer performs the following process:

First, the input text is split into tokens (i.e., units of text, such as words, subwords, or characters). Each token is then mapped to a numerical vector representation called token embedding, and positional encoding is added to each embedding to preserve information about the order of tokens in the sequence. The transformed embeddings, called token embeddings, pass through multiple Transformer blocks, each consisting of a multi-head self-attention mechanism followed by a multi-layer perceptron (MLP). The core innovation of the Transformer lies in the self-attention mechanism, which allows the model to process entire sequences in parallel and capture long-range dependencies more effectively than previous architectures. Within the self-attention mechanism, the model computes queries, keys, and values from the embeddings to determine how much each token should attend to each other in the sequence. Masking is applied to restrict attention to only earlier positions to maintain the autoregressive nature of the Transformer. Multiple attention heads run in parallel, enabling the model to capture different types of relationships across the sequence. Following self-attention, the MLP transforms the embeddings by applying non-linear transformations,

enabling the model to represent complex patterns. These operations are repeated across multiple Transformer blocks, where earlier blocks tend to capture low-level patterns and later blocks tend to capture high-level semantics. Finally, the transformed embeddings are projected into a probability distribution over possible next tokens. During inference, sampling hyperparameters such as temperature influence the next token prediction, enabling the model to iteratively generate text sequences.

2.2 *Traditional Learning Resources for Transformers*

Most resources for learning about Transformers are presented in traditional formats such as blog posts and video tutorials. A popular blog post is *The Illustrated Transformer* by Jay Alammar [2], which offers a detailed explanation of each model component using vector and matrix calculations. However, the information is presented in a lengthy format on a single webpage. While the depth can be beneficial, the article lacks a high-level overview of how all the components interact, which can be challenging for learners who are looking for a more end-to-end understanding of the Transformer. Video tutorials offer another approach, with popular examples including 3Blue1Brown’s video explaining the GPT model [1] and Andrej Karpathy’s video on implementing the GPT model [22]. However, these videos tend to focus on either mathematical details with formulas or implementation details with model code. These traditional formats often struggle to convey the dynamic processes within the Transformer [12], limiting opportunities to benefit from interactive learning methods that have been shown to improve learner understanding and engagement [26]. These limitations highlight a need for educational tools that support interactivity and dynamic feedback, enabling learners to engage more directly with the model.

2.3 *Interactivity in Learning Resources for Deep Learning Education*

Interactivity has been found to be an effective method for learning, even in the context of deep learning. Interactive articles have grown in popularity [44, 29, 7, 9], inspired by early work such as Chris Olah’s blog post [28], where learners could adjust model parameters and observe live feedback on their impact. Journals like Distill also embraced this format, combining narrative explanations with dynamic visualizations [17, 15]. Research has shown that such interactivity not only deepens conceptual understanding but also improves user experience and engagement by providing real-time feedback [44]. However, these interactive articles typically follow a predefined structure, which can limit exploratory learning.

To address these limitations, interactive visual tools have been developed to teach deep learning concepts. Two notable tools are CNN Explainer [40] and Diffusion Explainer [23],

which are web-based tools that enable users to interactively explore deep learning architectures. CNN Explainer offers multiple views that visualize the CNN architecture across different levels of abstraction, from high-level model overview to low-level mathematical operations with animations. A user study showed that learners not only gained a deeper understanding but also reported higher engagement and satisfaction compared to traditional learning methods [40]. Similarly, Diffusion Explainer explains diffusion models through interactive visualizations. Users can manipulate input prompts and model parameters to observe how these changes impact the generated outputs. A user study further highlighted its effectiveness, with learners finding the tool more understandable and engaging than traditional text-based resources [23]. Both tools provide real-time manipulation and feedback, offering a hands-on learning experience that enhances both understanding and engagement.

Despite these successes, interactive visual tools designed for teaching Transformers remain limited. To our knowledge, the only existing resources are LLM Visualizer [6] and TransforLearn [14]. LLM visualizer provides a 3D visualization of the entire GPT model; while it faithfully represents the model architecture and provides a linear algebra view of the Transformer model, it lacks higher-level intuitions and is so large that the connection between the input and output can get lost. Another tool, TransforLearn, provides an interface for interactively exploring the Transformer components. However, its explanations rely on static descriptions and do not adapt to user input, making it difficult to track how embeddings are transformed through the system. Additionally, its emphasis on token embeddings rather than core mechanisms like self-attention limits its educational value. The tool is also not publicly deployed [14]. Transformer Explainer addresses all these limitations as the first online interactive visualization tool offering real-time interactivity for learning how Transformers work.

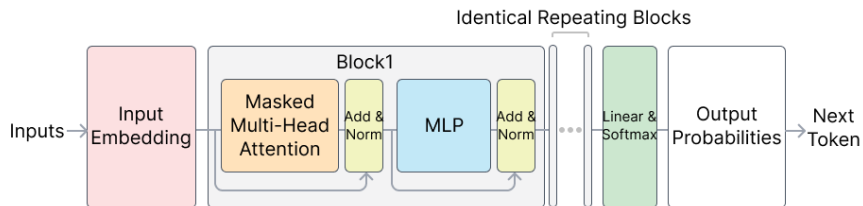


Figure 2: Diagram illustrating the data flow in a decoder-only Transformer architecture. Inputs are transformed into input embeddings, which are then passed through repeated identical Transformer blocks. Each block includes a masked multi-head self-attention, followed by a multi-layer perceptron (MLP), with residual connections (Add) and normalization (Norm). Finally, the output is passed through a linear layer and a softmax function to produce output probabilities for predicting the next token.

Adapted from original figure by Areee Cho.

3 Design Challenges & Goals

We outline the motivations behind our design choices for Transformer Explainer. In developing the tool, we identified four main design challenges and corresponding design goals to address them:

3.1 Unified View Preserving Data Flow Among Components

The Transformer architecture is dense, consisting of repeated blocks with many components such as multi-head self-attention and multi-layer perceptrons (MLPs) [13, 37]. This complexity can make it difficult for non-experts to understand the *data flow*—how inputs are processed and transformed through the model, as the data output from one component becomes input to the next (Fig. 2). Traditional visualization approaches often overwhelm users by presenting all details at once [6], or show components separately [14], which hinders learners from seeing how data moves through the model. However, research has shown that presenting all operations and their connections in a single view can help comprehension by showing how input data is progressively transformed into model predictions [40]. Therefore, we aim to create a connected visual summary of the Transformer architecture that emphasizes the progression of data transformations from input to model predictions. We would like to design an overview with a flow-focused representation. Inspired by recent studies that view Transformers as dynamic systems [11], we adapt the Sankey diagram—commonly used to depict flows—to illustrate how information moves through the Transformer architecture. This flow-based approach provides a visual summary that captures the continuous transformation of data as it passes through the model’s components and blocks. (Sec. 4.1)

3.2 Step-by-Step Visual Explanations of the Mathematics Behind Transformers

The complex mathematical operations within Transformers pose a challenge for learners [34]. Unlike architectures such as Convolutional Neural Networks (CNNs), which can leverage visual metaphors (e.g., filter sliding over images) [40], Transformers lack cognitive motivation [36] and rely on high-dimensional numerical operations that are not understandable on their own [27]. Operations performed on these representations—such as in the self-attention mechanism, where tokens simultaneously interact—further contribute to the complexity [37]. To address this challenge, we would like to adopt incremental, step-by-step visual explanations that gradually show each step of the computation leading to the output, inspired by educational research on algorithm visualization [12]. Our design reveals intermediate steps in the computation process—particularly within the self-attention mechanism—through interactive animations. By allowing users to hover over matrix elements and trace how values

are computed during operations such as matrix multiplication, our tool supports learners in constructing a mental model of the underlying mathematical operations [12], therefore improving understanding and engagement [21]. (Sec. 4.2)

3.3 Real-Time Experimentation Through Direct Manipulation of Sampling Hyperparameters

Many learners struggle to understand how Transformer models generate text—particularly how the output probability distribution over possible next tokens is used to sample the next token (Sec. 4.3), and how this sampling process is influenced by sampling hyperparameters that control prediction variability (Sec. 4.4). Educational resources tend to omit explanations of how this distribution is used during sampling and how sampling hyperparameters like temperature affect the outcome [6, 14]. As a result, non-experts may perceive the model’s behavior as magical or even attribute human-like characteristics to it [3]. To address this misconception, we would like to design an interactive interface that allows users to directly adjust the temperature hyperparameter in real time and immediately observe its impact on next token selection. By experimenting with different temperature settings, users can observe how higher values lead to more diverse outputs while lower values produce more deterministic results. This interactive experimentation helps demystify the generation process, reinforcing that the model’s behavior is not magic, but rather driven by a well-defined sequence of operations.

3.4 Web-Based Tool Powered by Live Model

Existing educational tools for Transformers rely on static explanations or offer limited interactivity [6], largely due to the technical difficulty of deploying deep learning models in real-time environments [32]. Hosting a live Transformer model in-browser is especially difficult due to its computational demands and the need for low-latency inference. As a result, current tools often use pre-selected examples [6, 23] or restricted outputs [40], limiting exploratory learning. To overcome these limitations, we developed a web-based tool that runs a live, fully functional Transformer model directly in the browser, enabling users to engage with the model directly without installations or specialized hardware. By open-sourcing our code, we also aim to support further development and research. (Sec. 4.6)

4 Visualization Interface of Transformer Explainer

4.1 Overview

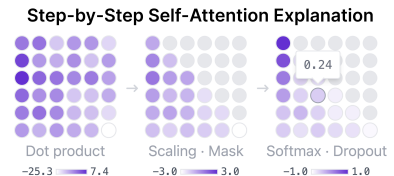
The visual design of Transformer Explainer (Fig. 1) draws inspiration from the Sankey diagram—commonly used to communicate the movement of data [31]—to present a high-level overview of how input flows through the Transformer components to generate next-token predictions. Gradient-colored paths illustrate the transformations of data (i.e., embeddings, which are illustrated by vertical bars) across model components, enabling an understanding of the Transformer architecture from a data-flow perspective. To avoid overwhelming users with the dense Transformer architecture, characterized by multiple blocks with many attention heads, the interface simplifies the visualization by displaying only one block and a single attention head at a time, reducing cognitive overload. Repeated Transformer blocks are suggested through gradually fading flow lines between blocks, implying continuity. Attention heads within a block are depicted as stacked cards, reinforcing the idea of repetition.

4.2 Step-by-Step Expanded Explanations

We developed interactive *Step-by-Step Detailed Explanation Views* (*Expanded Views*) to help users understand the mathematical operations within model components. Rather than overwhelming users by presenting all details at once, we enable users to activate these Expanded Views by each component’s title. Upon activation, a smooth animation gently expands the view, fading out surrounding areas while preserving overall context. This design enables users to focus on detailed internal operations without losing track of the broader context.

Self-Attention. The *Expanded Self-Attention View* visualizes the attention mechanism into three animated steps, enabling users to follow how values are transformed into final attention scores. This step-by-step explanation not only clarifies the underlying math but also enables users to compare intermediate results, helping reduce cognitive effort when tracking changes across the process [20].

Upon activation, the flows from key and query come together to perform a dot product, generating an intermediate matrix. This intermediate result is then duplicated and presented next to the original matrix to illustrate the scaling and masking operations. A similar approach is used to demonstrate the subsequent softmax and dropout steps. All matrix values are displayed as a heatmap using a purple color scale (e.g., -1.0 to 1.0). Users are able to interactively inspect numerical values of individual matrix elements to reveal numerical values through tooltips.

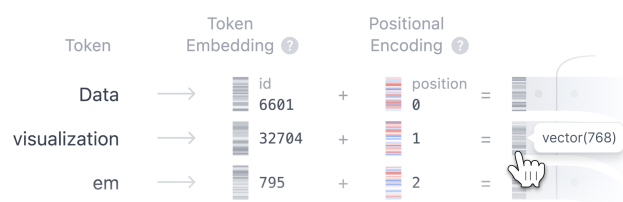


Probabilities. The *Expanded Probabilities View* guides users through the next-token prediction process by incrementally revealing each step in the underlying computation. When a user clicks to expand the view, the initial display of final token probabilities—ranked from most to least likely—unfolds into a step-by-step sequence: starting from logits, to their exponentiated values, and finally to normalized probabilities using softmax.



Embedding. The *Expanded Embedding View* shows how input tokens are transformed into numerical representations that feed into the model. When expanded, this view shows each token’s ID and its position in the sequence, along with the two vectors used to represent it (i.e., a token embedding and a position encoding). These vectors are added together to create the input embedding that is passed to the next model component. Each vector is displayed as a one-dimensional heatmap, and users can hover over it to see their exact dimensions.

Step-by-Step Embedding Explanation



Animated Matrix Multiplications.

In Transformers, embeddings are repeatedly transformed through matrix multiplications with pretrained weight matrices, changing their dimensions. These transformations are reflected in the Sankey diagram paths. Because such operations appear repeatedly throughout the model, they are visualized in interactive popovers: when a user clicks on a data path between embeddings, an animated popover opens, showing how input vectors are multiplied by weight matrices to produce a new matrix. Given the high dimensionality, embeddings and weights appear as simplified heatmaps, supplemented by numeric dimension labels provided below. Hovering over any output element highlights the corresponding input values involved in the computation, helping to clarify calculation details.



4.3 Real-Time Inference for Next-Token Prediction

Users can input custom text and observe in real time how the model predicts the next token by clicking [Generate](#). As soon as text is entered, it is tokenized and reflected in the visualization with a smooth animation that illustrates how tokens move through the model—showing changes in position, size, shape, and color of data flow elements—guiding users in understanding how the data evolves [19]. During this animation, a loading symbol appears within the input bar, and once the data flow reaches the output probability distribution, the predicted token is added. Users can continue clicking *Generate* to extend the sequence, gaining a visual understanding of how a Transformer constructs sentences one token at a time. To ensure usability in environments with latency or bandwidth constraints, the interface offers five example prompts preloaded with recalculated data, enabling immediate exploration without loading delays.

Examples ▾	Data visualization is ...
Examples ▾	Data visualization is useful

4.4 Adjustable Sampling Hyperparameter Influencing Next-Token Prediction

Transformer Explainer allows users to adjust the *temperature* sampling hyperparameter and instantly observe how changes impact the next-token prediction (Fig. 3). A slider next to the *Generate* button enables users to modify the temperature value, which controls the sharpness of the model’s output probability distribution. Lower temperatures produce sharper (more deterministic) distributions, while higher temperatures produce smoother (more random) ones. This interactive, real-time feedback helps users test their hypothesis and understand how Transformers make predictions—i.e., through probabilistic decisions, not through random guesses or “magic.”

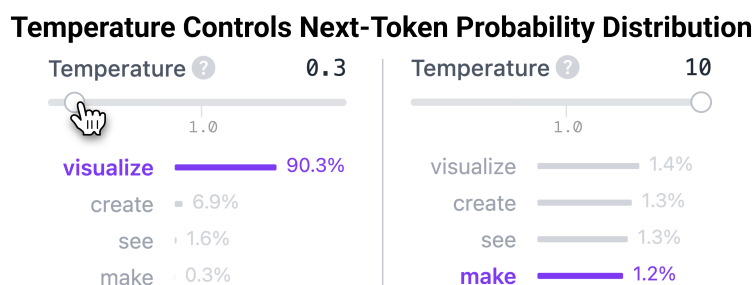
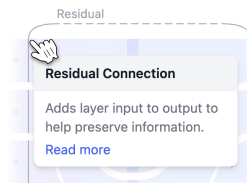


Figure 3: The temperature slider lets users interactively experiment with the temperature parameter’s impact on the next token’s probability distribution. **Left:** lower temperatures sharpen the distribution, making outputs more predictable. **Right:** higher temperatures smooth the distribution, resulting in less predictable outputs.

Figure and caption reproduced from [8].

4.5 Advanced Architectural Features

Our tool simplifies the visual representation of advanced architectural features, as understanding these elements is not required to follow the model’s core logic and data flow. To not overwhelm users with details, we use subtle visual cues—dots  indicate layer normalization, dropout, and activations, while lines  indicate residual connections. When users hover over a residual connection (see figure on the right), it transforms into a dashed, animated line that shows the flow from its source to its target. Hovering over any of these symbols also reveals a brief tooltip explanation, and users can click the *Read More* button to view a supplementary in-depth article just below the visualization.



4.6 Web-Based, Open-Source Implementation

Transformer Explainer is a web-based, open-sourced tool designed to help non-experts understand how Transformers work. It runs entirely within the browser—with no installation or specialized hardware necessary. We use a GPT-2 Small model from NanoGPT [22], built on HuggingFace Transformers [41], to provide model outputs for real-time experimentation. To run a live model directly in the browser, we converted the model from PyTorch to ONNX format and used the Onnx Runtime Web API [10] for inference. For improved performance, the model is split into smaller chunks that are downloaded in parallel and cached locally using IndexedDB, so users only need to download the model once. The frontend is built with Svelte [18] and D3.js [4], and the entire tool—including both the static state and the model—is hosted on GitHub Pages at <https://poloclub.github.io/transformer-explainer/>.

5 User Study

To assess how effectively Transformer Explainer supports users in learning about Transformers, we conducted a user study using a within-subjects design [33], in which participants compared our tool against another publicly available learning resource.

5.1 Procedure

Participants were recruited through Prolific, an online research platform. After providing informed consent, they completed an initial questionnaire about their familiarity with and interest in AI and text-generative AI, using a 5-point scale. Only participants with limited familiarity (rated as *1: Don't know what it is*, *2: Heard of it only*, or *3: Occasional users of AI-powered tools*) were eligible. Each participant interacted with both Transformer Explainer and a widely referenced blog post explaining decoder-only Transformers (GPT-2), identified through a top Google search result. To minimize ordering bias, the order in which participants interacted with the two resources was alternated across participants. Following each interaction, participants completed a short quiz to confirm meaningful engagement with the material. The quiz questions were easy for participants who used the tools (e.g., “Which of these features is NOT covered by Transformer Explainer?”). Out of the 250 initial participants, 145 met the eligibility criteria and correctly answered at least two out of three quiz questions for each resource.

Participants reported moderate familiarity with AI (average score of 2.86 for AI generally and 2.84 for text-generative AI specifically) and came from diverse professional backgrounds (e.g., finance, healthcare, retail, information services, manufacturing, education, construction, arts, public administration, and real estate). Feedback was collected using 5-point Likert-scale ratings focused on three areas: usability (7 questions), understanding of Transformers (9 questions), and the usefulness of specific features (8 questions, asked only for Transformer Explainer). At the end, participants indicated their overall resource preference. The entire session lasted around 40 minutes per participant, who received compensation of \$100.00.

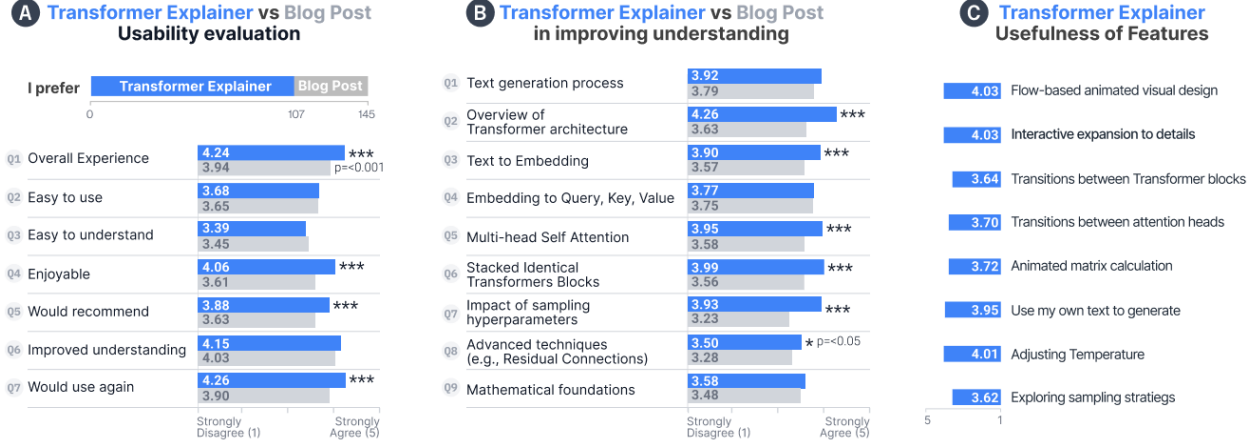


Figure 4: Average ratings from a 145-participant study comparing Transformer Explainer to a traditional blog post: **(A) Usability**: Most participants favored Transformer Explainer, rating it significantly higher in overall experience, enjoyability, and their likelihood of recommending or reusing it; **(B) Understanding**: Transformer Explainer was significantly more effective at improving participants’ understanding of the Transformer architecture and its underlying operations; **(C) Feature Usefulness**: Every feature in Transformer Explainer received higher ratings for usefulness. Statistical significance was assessed using a paired t-test ($p < .05 = *$, $p < .001 = ***$).

Figure by Aeree Cho.

5.2 Results and Design Lessons

We conducted paired t-tests to statistically evaluate participants’ ratings. Our findings revealed a clear preference for Transformer Explainer, with about 74% of participants preferring it over the static blog post. Specifically:

- **Improved usability**: Participants rated Transformer explainer higher on six out of seven usability metrics, with comparable ratings for ‘Easy to understand’ (Fig. 4A).
- **Improved understanding**: Participants consistently rated Transformer Explainer higher across all nine understanding metrics (Fig. 4B). Significant improvements were observed in understanding key concepts, such as the overall Transformer architecture, multi-head self-attention mechanism, and the effects of sampling hyperparameters.

Key insights from qualitative participant feedback included:

- **Interactive experimentation improves comprehension**: Participants particularly valued the ability to input their own text and immediately observe token-by-token predictions. This interactive experimentation helped deepen their understanding of model behavior and improved engagement. This feature also received high usefulness ratings (Fig. 4C).

- **Data flow visualizations clarify complex model architecture:** The tool’s data-flow-centric visualization was praised for making abstract concepts more concrete, improving participants’ mental models of the Transformer architecture. Participants reported significantly higher understanding of overall model architecture (Q2, Fig. 4B).
- **Real-time hyperparameter adjustments improve learning:** Users appreciated instant visual feedback from adjusting the temperature hyperparameter. This feature notably enhanced their understanding of probabilistic model behaviors (Q7, Fig. 4B).
- **Step-by-step expanded explanations simplify complex model operations:** Expanded explanations that incrementally reveal and visualize complex mechanisms, especially the self-attention mechanism, were highly effective in enhancing understanding of internal Transformer operations (Q3, Q5, Q7; Fig. 4B).
- **Interactive, playful experiences improve engagement:** The combination of interactive elements and visual animations created an engaging experience, like an “interactive museum exhibit,” making complex AI concepts more approachable and less intimidating. This was reflected in high ratings for overall user satisfaction (Q1), enjoyment (Q4), and willingness to revisit the tool for future study (Q7) (Fig. 4A).

5.3 Limitations

While participants provided largely positive feedback, they also identified certain usability challenges:

- **Interactivity increases complexity; onboarding can help:** Some participants noted an initial learning curve and suggested adding introductory materials, such as brief onboarding videos or guided walkthroughs, to ease initial interactions.
- **Support for deeper exploration:** Many users wanted to explore the attention mechanism in more depth—particularly how attention heads and blocks interact. Some requested additional features, such as the ability to adjust attention weights themselves.
- **Further breakdown of mathematical details:** Participants recommended enhancing visual mathematical explanations with additional textual annotations or explanatory popovers to better support non-expert learners.

Overall, the user study demonstrated Transformer Explainer’s effectiveness and user preference over traditional static learning resources, while also identifying opportunities for further enhancing its usability and educational value.

6 Discussion and Future Work

Extending Support to Other Modalities. Transformer Explainer currently supports text-based Transformers, but our user study revealed user interest in extending support to other modalities, such as multi-modal models like CLIP (for vision and language) and unimodal models like Vision Transformers (ViT) for vision and Whisper for speech [42]. Our visual design, based on flow-based approach and step-by-step expanded explanations, could adapt to these different architectures. Although, supporting tasks like image generation may require specialized techniques from prior work [23, 25] (e.g. dimensionality-reduced visual summaries for comparing generation trajectories or saliency maps for highlighting visual attention), presenting an interesting direction for future research.

Expanding Interpretability and Visualization of Attention. Our user study found that non-expert users are interested in more in-depth explanations of the attention mechanism in Transformers. This suggests that non-experts could benefit from more detailed explanations of the attention mechanism to deepen their understanding. Notably, most existing attention visualization tools [24, 38, 39, 43] have been developed primarily with expert users (i.e., researchers and practitioners) in mind. In future versions of our tool, we aim to thoughtfully adapt design elements from these advanced tools to help non-expert users explore how attention heads operate and interact across blocks—all while maintaining simplicity and usability.

Scaling to Larger Models. As Transformer models continue to evolve, models like GPT-4 are capable of processing larger inputs and capturing more nuanced contextual relationships. Our user study reflected a growing interest in educational tools that support the exploration of these modern, larger models. However, visualizing extended input sequences can quickly lead to visual complexity, potentially overwhelming non-experts. Furthermore, deploying such large models directly in the browser presents technical challenges [23]. Future work could investigate specialized visualization strategies in combination with optimized browser-based implementations (e.g., using WebAssembly [32] or model compression techniques) to make larger Transformer models more approachable to a broader audience.

7 Conclusion

We introduce Transformer Explainer, an interactive visualization tool designed to help non-experts learn how Transformers work in the context of text generation. Our tool combines a high-level, data flow-oriented overview with low-level, step-by-step explanations of mathematical operations. Users can interact directly with a live GPT-2 model in their browser, exploring in real time how different inputs and hyperparameters influence its behavior. Findings from our study show that Transformer Explainer improves users’ understanding of Transformers and their overall engagement.

8 Acknowledgments

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